

Sustainability or Stagnation? The Open Source Foundation Dilemma

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This essay examines whether the open source foundation model, responsible for much of the infrastructure we rely on, is well-positioned to thrive over the next decade. The essay reflects my perspective, informed by over two decades of experience working within the open source ecosystem.

Introduction

It has been stated that “open source has won” (I agree), and today, open source is the foundation on which most of our digital infrastructure is built, ranging from embedded systems to cloud, edge computing to AI. The “Open Source First” approach is now the default approach across nearly every domain and industry. To support this growing open source ecosystem, companies collectively invest hundreds of millions of dollars in open source foundations each year. These foundations coordinate governance, offer neutral project hosting, deliver operational support, and sustain developer communities. For the last twenty-five years or so, the model has been largely effective. I have been personally involved in it, first in the early 2000s at the Open Source Development Labs (the predecessor of the Linux Foundation), and then twice at the Linux Foundation, where, during my last tenure, I served as the Executive Director of LF AI & Data (6 years) and the founding ED of the PyTorch Foundation (1.5 years). For context, I’ve spent as much time at the Linux Foundation as I did at Volvo Cars, Samsung Research, HP/Palm, Motorola, and Ericsson Research combined.

The question I am debating in this essay is whether the open source foundation model is viable for the next decade.

I am certainly not predicting (or expecting) collapse. What I'm exploring is a hypothesis that the dominant support and funding models of today's open source foundations may be nearing the edge of long-term viability. This is not because they've failed or they have not previously delivered on their promise, but because the environment is changing and the ground is shifting under them.

The Case for Sustainability: The Current Model Will Hold

Starting with the arguments in support of the model's sustainability, we can agree that the current model has a lot going for it, especially from the perspective of enterprise funders or, in more practical terms, foundation members.

1. **Holding value proposition:** First, the core value proposition of the current model still holds. These foundations offer neutral governance, infrastructure support, ecosystem development, legal and licensing guidance, marketing and events support, and fundraising mechanisms, all of which are essential requirements when multiple competitors want to collaborate without creating any entanglements. This value proposition remains a compelling offer, particularly in high-risk domains such as cloud orchestration, security, cryptography, or AI.
2. **Foundation scaling model:** Many open source foundations have matured into hosting dozens or even hundreds of projects, and some have evolved into umbrella organizations. That setup creates efficiencies that no single company could replicate or support on its own. Many of the foundation-specific characteristics, such as shared infrastructure, centralized staff, and supporting ecosystem services, contribute to lowering operational friction and boosting visibility.
3. **Organizational inertia:** Third, there's organizational inertia to factor in, which is often overlooked. In many large companies, foundation membership is a normalized budget, typically included in developer relations, engineering, or strategy (CTO) budgets. The annual invoice gets paid not because it's scrutinized and approved, but because it's familiar. For some strategic-tier members, these six-figure payments are a rounding error compared to broader R&D expenditures. That institutional habit creates a kind of friction against withdrawal; leaving a foundation isn't just a financial decision, it's a political one, and conveys a particular perception that organizations want to avoid. For now, that built-in stickiness continues to benefit the existing model.
4. **Ability to pivot:** Fourth, these foundations have evolved. Kubernetes emerged to reshape infrastructure, and then CNCF emerged to hold and support the entire open source cloud-native ecosystem. Software supply chain attacks have become a significant concern, and the OpenSSF has

responded. Open source license compliance requires support; OpenChain and SPDX are born. AI is becoming critical; foundations have started onboarding projects and building communities around it. The foundation’s ability to pivot has historically kept the model relevant.

5. **Supporting human connections:** Finally, foundations continue to foster human communities, an often undervalued aspect. Beyond the services mentioned above, foundations provide mentorship, contributor onboarding, diversity initiatives, cultural stewardship, and are the home of a few fellows of critical projects. That soft infrastructure is easy to overlook but hard to replace.

The existing model continues to deliver governance, legitimacy, and shared infrastructure. Still, we need to ask whether that will be enough in a world being reshaped by AI, fragmented geopolitics, and new funding pressures.

Foundations like the Linux Foundation have demonstrated a real capacity to adapt, launching the OpenSSF in response to security threats, founding the LF AI & Data Foundation and the PyTorch Foundation for open source AI, and engaging with public-sector efforts within the scope of several other umbrellas and projects. But the scale and speed of the shifts ahead, from AI-native codebases to geopolitical funding divergence, will demand a level of structural change that goes beyond adding a new project or launching a new umbrella foundation.

The Case Against Sustainability: The Current Model is Under Strain

If you zoom out, you’ll see stress signals accumulating: funding concentration, AI-native disruption, rising ROI scrutiny, and geopolitical divergence. None of them is fatal alone, but together, they signal a model under pressure.

1. **Fragile Funding Concentration:** A small number of major companies shoulder most of the financial burden. If even a few reduce their support due to economic downturns, mergers, or internal reprioritization, the foundation’s fiscal base becomes shaky fast.
2. **ROI Pressure and Value Perception:** Companies are scrutinizing what they get in return. The blanket “ecosystem support” argument is becoming a tougher sell in performance-driven organizations. Increasingly, funding follows influence. Suppose projects aren’t directly aligned to product outcomes or market traction. In that case, executives hesitate to commit to higher-level membership, in addition to allocating resources, to ensure active engagement. An ongoing six-figure annual fee, in addition to the human resources required to support a proper level of participation, often demands tangible ROI, and that connection isn’t always clear. The logic of proximity (funding what we use and where we have influence) is starting to trump the logic of the collective good.

3. **The AI Disruption Curve:** Foundations were built on the assumption of human creativity, developers writing code, reviewing contributions, and collaborating in public. That assumption is now under stress. As large language models begin generating not only code snippets but also complete modules and patches, the definition of a “contributor” is evolving. For example, we’re already seeing AI-generated PRs in widely used repositories. This raises questions about code provenance, licensing (especially when models are trained on GPL/AGPL code), and the governance norms for AI-synthesized work. Many foundations lack clear policies for attribution, audit trails, or even basic contributor identity in these contexts. That gap will grow fast. On the other hand, AI-generated code could accelerate contributions, reduce maintenance effort, and improve onboarding. But to unlock that upside, foundations will need new governance primitives that handle code provenance, attribution, and hybrid workflows. This isn’t a crisis today, but it’s a fast-emerging pressure point. Foundations that proactively define attribution, audit, and governance norms for AI-generated contributions will shape the evolution of legitimacy and accountability. Those who delay will struggle to keep pace with a contributor base that’s shifting beneath them.
4. **Geopolitical Fragmentation:** Foundations built for global neutrality now face a hard test: can they serve members with diverging regulatory obligations and national priorities without losing legitimacy in either camp? 2025 has shown us that the world is going regional. China’s OpenAtom Foundation, Europe’s NGI, and the Sovereign Tech Agency, along with calls for digital sovereignty, are shaping open source engagement increasingly by national interests. These changes are even affecting the responsibilities and behaviors of Open Source Program Offices, reshaping their role to be strategic risk radars. This updated OSPO mission entails collaborating with legal teams, monitoring dependency health and jurisdiction, and exploring ways to make meaningful contributions to both international and domestic ecosystems amid ongoing geopolitical tensions and calls for tech sovereignty. Companies are under pressure to fund local initiatives. Open source foundations built for global neutrality may find that maintaining neutrality becomes harder when their member base is being pulled in different directions.
5. **Legal jurisdiction:** One increasingly visible stress point is legal jurisdiction. Global foundations must still operate under specific national legal regimes. This setup creates inconsistencies, such as when US sanctions policies affect contributor eligibility or project governance decisions, regardless of global sentiment. The expulsion of Russian contributors from the Linux kernel, following US sanctions, is one such example. While defensible under the foundation’s legal obligations, it highlighted a reality where “global” often implies “US-aligned” by default. This dissonance is encouraging the rise of explicitly regional foundations (OpenAtom in China or the Sovereign Tech Agency in Europe), which are aligned with

their legal and policy environments by design.

These pressures don't signal failure. They suggest that the current model is based on assumptions that may no longer be valid.

Where Do We Stand?

We're at a moment of reassessment. Foundations still deliver value, but the assumption that their funding and relevance will remain steady is no longer a safe bet. Companies that once renewed memberships reflexively are now asking sharper, ROI-driven questions. Some have downgraded sponsorships and/or shifted to project-level investments instead. At the same time, maintainers and core contributors are exploring alternatives outside the traditional foundation umbrella, including models such as GitHub Sponsors, Open Collective, sovereign community governance, and dedicated funding from universities and public sector initiatives. Take Mastodon or Home Assistant: both have matured with community-backed financial models that bypass large foundations entirely.

Regional ecosystems are also asserting themselves. In the EU, the Sovereign Tech Agency and NGI initiatives are funding critical open infrastructure with public money, while in China, the OpenAtom Foundation is positioning itself as the domestic alternative to US-centric foundations.

This isn't a collapse. It's strategic drift, and without intentional recalibration, the gravitational center of open collaboration may shift elsewhere. Some foundations are growing, but growth driven by legacy momentum or a narrow funding base isn't structural resilience. The risk here isn't implosion, it's strategic overconfidence in a model that may be eroding at its edges.

Alternative Models

We're no longer in a single-institution world. New models are emerging not to replace foundations, but to challenge their monopoly on legitimacy, funding, and community scaffolding.

1. Open Collective and GitHub Sponsors have enabled maintainers to collect donations directly from users and companies. These platforms offer transparency, lower overhead, and a closer link between contribution and compensation. However, they don't address governance, IP, long-term maintenance challenges, or ecosystem development, and aren't designed for large, multi-stakeholder projects.
2. The Sovereign Tech Agency in Germany is a public-sector-led initiative that supports critical open source infrastructure. Its strength lies in its targeted, mission-aligned support. It doesn't try to replace foundations but complements them with strategic injections of capital into high-impact areas.

3. We're also seeing emerging experiments with community-owned Decentralized Autonomous Organizations (DAOs), municipally funded open source, and hybrid consortia that combine foundation scaffolding with regional or domain-specific governance.

None of these models has yet replaced or will replace the foundation model, but they're chipping away at the idea that there's only one way to organize and sustain open source.

Foundations continue to play a significant role in shaping open source culture, from mentorship pipelines to global summits. But culture alone can't carry models that don't evolve structurally. As funding pressures shift and contributor identities change, we risk community scaffolding being misaligned with the new reality.

Furthermore, these emerging models typically lack the IP frameworks and antitrust protections that are central to why large enterprises engage with foundations in the first place. Foundations don't just provide IT infrastructure, event support, and governance, among many other services. They create a legally protected, vendor-neutral collaboration space that makes it safe for competitors to contribute to shared infrastructure without crossing regulatory lines. That role, often invisible, is challenging to replicate in lighter-weight models. As alternatives mature, duplicating or replacing these legal scaffolds will be a defining challenge.

To be clear: this isn't about tearing down the foundation model. It's about realizing or even accepting that in a pluralistic future, no single model will meet the needs of all. New experiments from DAOs to sovereign tech funding aren't threats. They are signals that legitimacy and structure can come in more forms than we've historically accepted.

What Might Recalibration Look Like?

If the current model needs to adapt, what does that mean? I don't know for a fact, but here are a few speculative, plausible shifts:

- **Pay-for-impact funding:** Companies fund foundations or projects based on metrics that reflect usage/development growth, or ecosystem contributions tied to business outcomes. This model better aligns incentives, but carries real risks. Metrics can mislead, especially in non-code domains where contribution signals are more complex to quantify. A one-size-fits-all metric system will fail to capture the diverse dynamics of individual projects. Regionally anchored institutions, such as the Sovereign Tech Agency or OpenAtom, may be better positioned to develop meaningful local impact measures grounded in legal and cultural context. The future of open source funding may not only be usage-based, but also pluralistic in how value is measured and justified.
- **Tiered governance rights:** Foundations could evolve to offer different

levels of governance participation based on contributions, not just financial contributions. That means seats at the table for major code contributors, critical downstream users, or community advocates, not just top-tier sponsors.

- **AI-native contributor frameworks:** As code synthesis tools mature, foundations will need policies for identifying/tracking non-human contributors, reviewing generated code, and sharing legal responsibility. Ignoring this transition risks irrelevance.
- **Regional forks with global bridges:** Some foundations may need to formalize localized substructures that comply with national funding rules and digital sovereignty policies, while still contributing upstream and preserving interoperability through a shared global layer. Any credible alternative to the current model will need to offer a legal safe harbor for IP pooling, cross-company governance, and antitrust-safe collaboration, features that are deeply embedded in the foundation structure today.

These aren't mutually exclusive models, and they serve as a good starting point for conversation and experimentation.

The future is likely to involve a combination of these approaches and other potential methods, which will be tested and refined through experimentation.

Each of these approaches carries tradeoffs: more complexity, governance friction, or legal uncertainty. But standing still isn't risk-free either. Recalibration is no longer optional for the current model, and it's how we future-proof collective infrastructure. If your company invests in open source through foundations, you are likely to undergo a review of your foundation strategy. What's the intent behind your funding? What kind of impact, control, or stewardship are you supporting? How do these align with your product, policy, or engineering priorities?

Foundations are and remain useful. The question that is open for debate now is whether the structure that worked for the first 25 years of "industrial" open source can carry us into an AI-native, geopolitically fractured, impact-scrutinized future. That future will require a broader toolkit that includes foundation infrastructure, public-sector funds, regional hosts, flexible governance models, and purpose-built legal scaffolding. Some open source projects and initiatives will thrive in classic foundations. Others may split, localize, or evolve entirely new forms.

The next phase of open source sustainability will not evolve around a single model, but rather about growing a diversified portfolio of legitimacy, impact, and governance options.

About the Author

Dr Ibrahim Haddad is currently Head of Infotainment at Volvo Cars. He is widely recognised for his leadership in building and scaling Open Source Program Offices (OSPOs), having founded and directed them across multiple organisations, and advised both corporations and NGOs on open source strategy and governance. Previously, he served as Vice President of Strategic Programs (AI) at the Linux Foundation, where he was the Executive Director of LF AI & Data and the Founding Executive Director of the PyTorch Foundation. During his last tenure at the Linux Foundation, his core mandate was to support and grow the open source AI ecosystem by providing a neutral, trusted hub for developers and organizations to code, manage, and scale open source AI projects. Before the Linux Foundation, Dr Haddad served as Vice President of R&D and Distinguished Engineer at Samsung Research Silicon Valley. His career includes R&D technical and leadership roles at Ericsson Research, Open Source Development Labs, Motorola, Palm, and Hewlett-Packard. He holds a PhD with honours in Computer Science from Concordia University (Montréal, Canada).

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